

Instructions for installation, operation and maintenance of series type **R2-xxxx/y**

according to ČSN EN 61439-1 ed. 3, ČSN EN 61439-2 ed. 3

General description:

PSC switchboards designed for the protection of electric lines for buildings installed by the producer according to requirements of the customer, the project documentation and in accordance with the respective provisions of ČSN EN 61439-1 ed. 3 in switchboard boxes that are a combination of one or more switching low-voltage devices with secondary control, measurement, signaling, regulation and protective equipment, including all internal electric joints, mechanical connections and construction parts. For demands to ensure compatibility with the nominal values of circuits to which the switchboard is to be connected and with the conditions for electrical installation, the form is used with the characteristics of the specific switchboard which is an integral part of this manual. Switchboards in this series are designed for operation by users with an electrical technical qualification – Section 5 according to Government Regulation 194/2022 Coll., as amended. In addition to the switchboard, the user will receive the accompanying documentation in the Czech version, which contains:

- characteristics of the specific switchboard
- these instructions for installation, operation and maintenance,
- connection diagram,
- document on piece verification of the specific switchboard
- EU declaration of conformity
- manuals for use and maintenance from each manufacturer of the components installed in the switchboard.

Storage

Do not store the switchboard in unacceptable conditions, in particular, protect it against excessive moisture (e.g. damp premises which are not finished in terms of construction), high temperatures and frost (outside the range -25 °C to +40°C). Prevent any damage to the product.

Transport

Switchboards can be transported in usual, adequately fixed positions. Secure the wall by fitting control and signalling elements against damage. Protect against excessive moisture, temperature, frost, dust, etc., prevent any falling. When transporting, do not expose the switchboard to excessive shocks and handle the switchboard with maximum care. Heavier switchboards are fitted with aids for use with mechanization. Manipulation with switchboards using lifting mechanisms must only be performed by a person with valid card for operating lifting mechanisms.

Installation, operation and maintenance:

- 1) With respect to the character of the product and any threat, the following instructions and generally valid safety regulations must be kept.
- 2) During switchboard installation and maintenance, respect the characteristic data for the specific switchboard. To ensure compatibility with the nominal values of the circuits to which the switchboard is to be connected and with the conditions for electrical installation in the location of the switchboard, the form is used with the characteristics of the specific switchboard, which is an integral part of this manual.
- 3) The connection of the switchboard to the source of voltage in the place of installation, connection of output circuits, maintenance and repairs must only be done by a fully qualified employee – Section 5 according to Government Regulation 194/2022 Coll., as amended.
- 4) Operation must only be by an employee – Section 4 according to Government Regulation 194/2022 Coll., as amended.
- 5) Before installation, check the condition of the switchboard, particularly that it was not damaged during transport or release of electric components. A record is to be entered into the assembly diary.
- 6) During the installation of boxes for built-in switchboards, check for any deformation (e.g. expansion of the inside of the box).
- 7) During installation and connection of the supply line and output lines, do not break the declared coverage and connecting terminals and that the surface paths and air distances are not reduced. When connecting the line into the screwed terminals, tighten the joints by the torque wrench with the selected torque moment according to the joining method and the recommendation of the producers (e.g. LTN circuit breaker terminals max. 3.5 Nm).
- 8) During the installation of switchboards with protective measures for double insulation, do not breach the principles for protective measures according to ČSN 33 2000-4-41 ed. 3. During the installation of switchboards with protective measures for automatic disconnection from the source, pay attention to the grounding and connection of the protective conductor according to ČSN 33 2000-4-41 ed. 3.

- 9) After termination of the installation of the switchboard and the connection of the output lines to the switchboard terminals before commissioning, inspect the settings for the required values of protection against overloading according to the label values for connection of equipment, the protective measures, in particular the tightening of the conductor joints for the protective circuit and consequently conduct an initial revision of the connection of the switchboard for installation in accordance with Government Regulation 190/2022 Coll., as amended..
- 10) Do not make any modifications to the internal connections of the switchboard that are not approved by the producer.
- 11) After modifications to the internal connections in the switchboard approved by the producer, proceed in the same manner as in the case of the connection of a new switchboard.
- 12) Replace the damaged devices with the same devices with identical parameters. After repair, make a written record with the signature of the qualified employee who carried out the repair pursuant to ČSN 33 1500 Article 2.7.
- 13) Before a qualified employee starts to work with the electrical equipment, follow the provisions of ČSN EN 50110-1 ed.2, and disconnect the electrical equipment from the electricity network, secure the workplace and check that there is no hazardous voltage from any other source. Pay attention to backup sources (UPS), motor-generators, condensers, induction, etc.!
- 14) The warning only applies to Class A switchboards produced for industrial environment EMI:

This product was designed for environment A. Use of this product in environment B may cause undesired electromagnetic interference. In this case, the user may be required to take reasonable measures to mitigate this interference.

- 15) Maintenance by a qualified employee is usual and only requires occasional inspection of the switchboards (inspection of mechanical damage of covers, insulation and labelling).
- 16) If fuses are installed in the switchboard, then the specified types and rated values of fusible links are listed in the List of Fuses and in the attached manufacturing documentation.
- 17) Regularly revise switchboards within deadlines according to the environment in which they are installed. During revision, clean internal areas and tighten the bolt joints.
- 18) If the equipment is damaged or is not correctly functioning, it must be immediately disconnected from the supply and then call the manufacturer of the switchboard or call an employee with the minimum qualification in accordance with Section 5 of Government Regulation 194/2022 Coll., as amended.

Disconnection from operation

After the expiration of the service life of the switchboard, it must be ecologically liquidated. Due to the expected long life of the switchboard, it is necessary to respect the following valid regulations.

Guarantee and claims

The producer provides the guarantee for production and hidden defects for a period of 24 months from the date of sale, unless otherwise stated in the contract or order.

For the application of a claim to the product, a defect or non-functionality, then:

- document the purchase and identification of the claimed product (e.g. type and serial number),
- submit documents about the correct installation and maintenance of the switchboard (e.g. revision report on the correct installation and connection of output circuits of the switchboard, records of regular inspections),
- state the purchase date,
- describe the defect,
- state who is making the claim,
- list the names and addresses and other contacts of the claimant,

Under no circumstances will the seller accept any claim for defects occurring through unqualified servicing, unprofessional handling, use and installations which are in variance with the instructions contained in this manual and generally binding legal regulations. The guarantee does not apply to any damage due to natural disasters, installation of the switchboard outside the range of the stated working conditions, damage by force or in cases or unauthorized interference with the product.